Among-population disequilibrium between unlinked ancestry clusters is genome-wide, not pair-specific

Draft results summary (Module D, the intrinsic arm). Methods are deferred to Supplementary Materials. Findings are descriptive: the screen characterises the observed distribution of among-population two-locus disequilibrium and ranks candidate pairs, but a pair-specific incompatibility is licensed only as an excess over the forthcoming recombination-matched neutral null. All statements below concern that observed distribution.

Results

Design. An intrinsic incompatibility between two loci acts on their joint parental ancestry, and across independent replicate populations should leave a pair-level trace that per-locus sorting cannot show: the two loci fixing compatible ancestry in the same direction repeatedly, so that their allele frequencies covary across populations. For loci on different chromosomes this covariance cannot be produced by linkage, and under drift alone independent replicates sort unlinked loci independently, so its neutral expectation is near zero. We therefore screened all pairs of *unlinked* (different-chromosome) LD clusters that were polymorphic in the parents (pooled parental MAF ≥ 0.15; 27,223 of the 32,840 consensus clusters), a little over 3.5×10^8 pairs. The among-population covariance was measured as R_{st}, the correlation across the twenty populations of a cluster's mean consensus dosage; the exact Ohta decomposition of two-locus disequilibrium into within- and among-population components was then evaluated on the 387,841 unlinked pairs reaching $|R_{\mathrm{st}}| \geq 0.7$. Same-chromosome (linked) pairs, where linkage is present, serve as an internal reference.

Unlinked pairs carry as much among-population disequilibrium as linked pairs. If elevated among-population disequilibrium between unlinked clusters reflected specific interactions, unlinked pairs would show a heavier tail of R_{st} than linked pairs, where the same tail is expected from linkage alone. They do not. The fraction of pairs exceeding $R_{\mathrm{st}} = 0.7$ is essentially the same for the two classes—0.068% of unlinked versus 0.083% of linked pairs, a ratio of only 1.2—and the two R_{st} distributions are superimposed across their whole range (1.1× at $R_{\mathrm{st}} \geq 0.5$). Physical linkage barely raises among-population disequilibrium above the unlinked background, which means that background is not a property of particular pairs but a genome-wide one.

The decomposition points to structure, not epistasis. On the high-R_{st} pairs the Ohta decomposition is unambiguous about the source of the disequilibrium. The among-population component exceeds the within-population component in virtually every pair (median $D^2_{ST} = 0.13$ versus $D^2_{IS} = 0.005$; $D^2_{ST} > D^2_{IS}$ in $> 99.99\%$ of pairs), while the disequilibrium of the averaged population is negligible (median $D'^2_{ST} = 0.005$, below D'^2_{IS} in every pair). A large among-population variance component together with a vanishing average disequilibrium is the classic signature of correlated allele-frequency *divergence* among subpopulations— drift and shared structure—rather than of a systematic two-locus association of the kind epistatic selection would maintain within populations.

A single shared ancestry axis accounts for the pattern. The covariance is predominantly positive: 62% of the high-R_{st} pairs covary in the same direction, and among pairs in which both clusters are themselves directionally sorted, same-direction pairs outnumber opposite-direction pairs 1.3 to 1. This is what a single genome-wide ancestry gradient produces— populations that fixed

F. aquilonia ancestry did so across many diagnostic loci at once, so those loci covary positively regardless of any interaction between them. The among-population disequilibrium between unlinked clusters is thus dominated by the same directional ancestry sorting characterised in the per-locus analyses, now seen as the covariance it induces between loci, rather than by pair-specific coupling.

Interpretation and outlook. Before a neutral null, the screen finds no evidence of pair-specific incompatibility: among-population two-locus disequilibrium between unlinked clusters is abundant but is statistically indistinguishable from the linked reference and reads, by its own decomposition, as correlated ancestry divergence along one genome-wide axis. This is the expected descriptive outcome, and it sharpens rather than answers the intrinsic question. A Dobzhansky–Muller interaction would appear as an *excess* of among-population disequilibrium, at specific unlinked pairs, over what neutral sorting of the same replicate populations produces. Module D is therefore run as a distribution-generator: it emits the observed R_{st} and among-population disequilibrium distributions, and candidate incompatibilities are defined downstream as the pairs exceeding the recombination-matched neutral null, which is evaluated by applying the identical pre-filter and decomposition to simulated genotypes. That comparison, with the concurrent linked-pair reference, is the decisive next step.